

WESTERN SYDNEY UNIVERSITY



SCHOOL OF BUSINESS

ASSIGNMENT COVER SHEET

STUDENT DETAILS

Student name: Nguyễn Minh Nguyệt

Student ID number: 22218252

UNIT AND TUTORIAL DETAILS

Unit name: Analytics Programming

Unit number: COMP1013

Tutorial group: _____

Tutorial day and time: 12:00 – 15:00 Sun

Lecturer or Tutor name: Assoc. Prof. Nguyễn Tấn Lũy

ASSIGNMENT DETAILS

Title: Individual Assignment

Length: 2,000 words

Due date: 26/07/2026

Date submitted: 26/07/2026

Home campus (where you are enrolled): Vietnam Campus

DECLARATION

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Individual Report

Nguyễn Minh Nguyệt - 22218252

2026-07-25

Data Preparation:

```
# Read data
```

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages —
```

```
tidyverse 2.0.0 —
```

```
## ✓ dplyr      1.1.4      ✓ readr      2.1.6
```

```
## ✓ forcats   1.0.1      ✓ stringr    1.6.0
```

```
## ✓ ggplot2    4.0.1      ✓ tibble     3.3.0
```

```
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.2
```

```
## ✓ purrr      1.2.0
```

```
## — Conflicts —
```

```
tidyverse_conflicts() —
```

```
## ✖ dplyr::filter() masks stats::filter()
```

```
## ✖ dplyr::lag()      masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all  
conflicts to become errors
```

```
library(scales)
```

```
##
```

```
## Attaching package: 'scales'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##      discard
```

```
##
```

```
## The following object is masked from 'package:readr':
```

```
##
```

```
##      col_factor
```

```
matches_raw <- read_csv("Matches.csv")
```

```
## Rows: 964 Columns: 16
```

```
## — Column specification
```

```
## Delimiter: ","
```

```
## chr  (11): MatchID, TournamentID, MatchName, Stage, MatchDate, StadiumID, Ho...
```

```
## dbl  (4): HomeTeamScore, AwayTeamScore, ExtraTime, PenaltyShootout
```

```
## time (1): MatchTime
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
stadiums <- read_csv("Stadiums.csv")
```

```
## Rows: 193 Columns: 4
```

```
## — Column specification
```

```
## Delimiter: ","
```

```
## chr (4): StadiumID, StadiumName, City, Country
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
teams <- read_csv("Teams.csv")
```

```
## Rows: 85 Columns: 3
```

```
## — Column specification
```

```
## Delimiter: ","
```

```
## chr (3): TeamID, TeamName, TeamCode
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
tournaments <- read_csv("Tournaments.csv")
```

```
## Rows: 23 Columns: 2
```

```
## — Column specification
```

```
## Delimiter: ","
```

```
## chr (2): TournamentID, TournamentName
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Merge dataset
```

```
matches_joined <- matches_raw %>%
```

```
  left_join(stadiums, by = "StadiumID") %>%
```

```
  left_join(tournaments, by = "TournamentID")
```

```
# test
```

```
glimpse(matches_joined)
```

```
## Rows: 964
```

```
## Columns: 20
```

```
## $ MatchID      <chr> "M-1930-01", "M-1930-02", "M-1930-03",  
"M-1930-04", "M...
```

```

## $ TournamentID    <chr> "WC-1930", "WC-1930", "WC-1930", "WC-1930",
"WC-1930",...
## $ MatchName       <chr> "France v Mexico", "United States v Belgium",
"Yugosla...
## $ Stage           <chr> "group stage", "group stage", "group stage",
"group st...
## $ MatchDate       <chr> "7/13/1930", "7/13/1930", "7/14/1930",
"7/14/1930", "7...
## $ MatchTime       <time> 15:00:00, 15:00:00, 12:45:00, 14:50:00, 16:00:00,
14:...
## $ StadiumID       <chr> "S-193", "S-192", "S-192", "S-193", "S-192",
"S-192", ...
## $ HomeTeamID      <chr> "T-28", "T-80", "T-84", "T-59", "T-03", "T-13",
"T-84"...
## $ AwayTeamID      <chr> "T-44", "T-06", "T-09", "T-54", "T-28", "T-44",
"T-07"...
## $ HomeTeamScore   <dbl> 4, 3, 2, 3, 1, 3, 4, 3, 1, 1, 6, 4, 1, 4, 3, 6, 6,
4, ...
## $ AwayTeamScore   <dbl> 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 3, 0, 0, 0, 1, 1, 1,
2, ...
## $ ExtraTime       <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, ...
## $ PenaltyShootout <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, ...
## $ HomePenalty     <chr> "?", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0",
"0",...
## $ AwayPenalty     <chr> "0", "0", "0", "?", "0", "0", "0", "0", "0", "0", "0",
"0",...
## $ Result          <chr> "home team win", "home team win", "home team win",
"ho...
## $ StadiumName     <chr> "Estadio Pocitos", "Estadio Gran Parque Central",
"Est...
## $ City            <chr> "Montevideo", "Montevideo", "Montevideo",
"Montevideo"...
## $ Country         <chr> "Uruguay", "Uruguay", "Uruguay", "Uruguay",
"Uruguay",...
## $ TournamentName  <chr> "1930 FIFA World Cup", "1930 FIFA World Cup",
"1930 FI...

```

Question 1: Data Cleaning and Formatting

```
matches_clean <- matches_joined %>%
  # Replace "?" with NA in all character columns
  mutate(across(where(is.character), ~ na_if(.x, "?"))) %>%
  # Convert specific columns to factors
  mutate(
    Result = as.factor(Result),
    Stage = as.factor(Stage),
    Country = as.factor(Country),
    ExtraTime = as.factor(ExtraTime)
  ) %>%
  # Convert penalties to numeric and replace NAs with 0
  mutate(
    HomePenalty = as.numeric(HomePenalty),
    AwayPenalty = as.numeric(AwayPenalty),
    HomePenalty = replace_na(HomePenalty, 0),
    AwayPenalty = replace_na(AwayPenalty, 0)
  )
# Test
summary(matches_clean %>% select(Result, Stage, Country, ExtraTime,
HomePenalty))
```

##	Result	Stage	Country	ExtraTime	
##	away team win:240	group stage	:676	Germany:102	0:891
##	draw :179	round of 16	: 97	Brazil : 86	1: 73
##	home team win:545	quarter-finals	: 70	Mexico : 84	
##		semi-finals	: 38	France : 82	
##		second group stage:	36	Italy : 69	
##		final	: 21	Qatar : 64	
##		(Other)	: 26	(Other):477	
##	HomePenalty				
##	Min. :0.0000				
##	1st Qu.:0.0000				
##	Median :0.0000				
##	Mean :0.1048				
##	3rd Qu.:0.0000				
##	Max. :5.0000				
##					

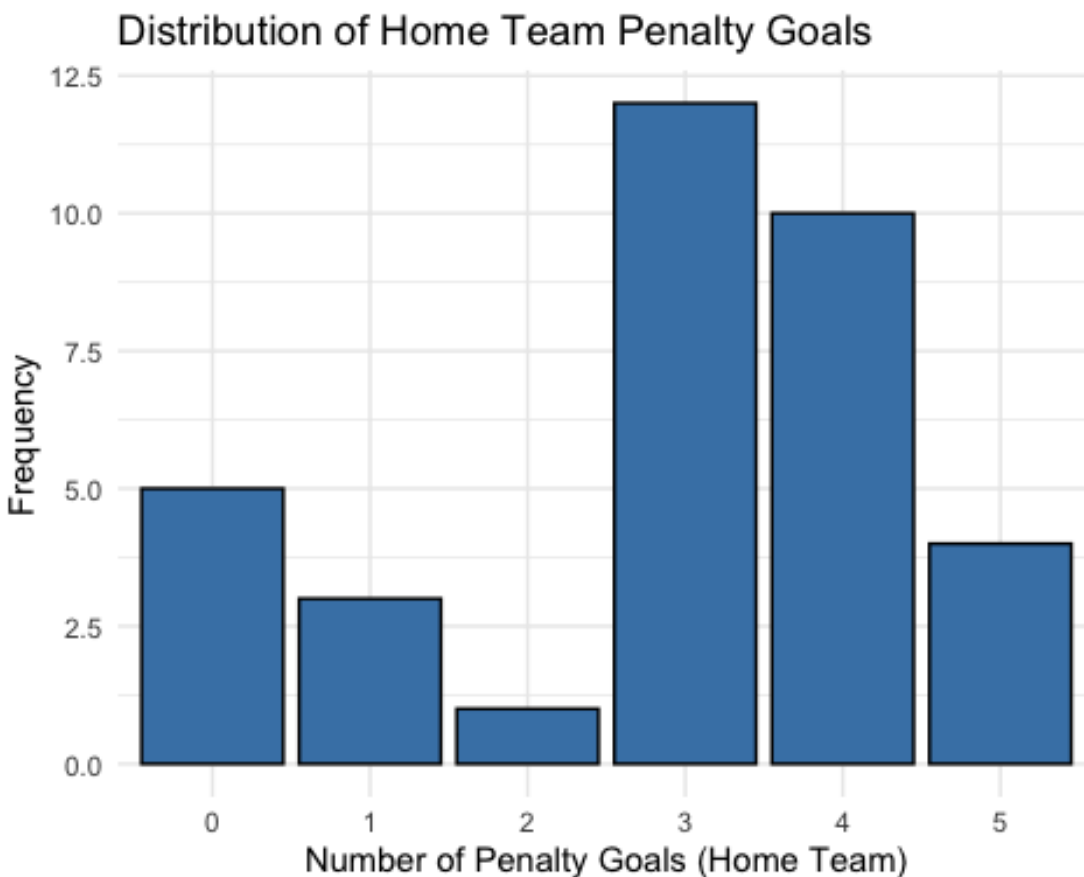
This algorithm was chosen to address the problem of missing data and the conversion of different data types due to the efficiency of the functions used: `na_if`, `as.factor`, and `replace_na`. The data distribution regarding the number of penalty goals scored at home is illustrated by a `geom_bar` chart, as the number of goals is a discrete variable. In the `geom_bar` chart, the frequency of each type of goal and the overall trend of the data can be observed.

```

# Visualization for Q1
penalty_matches <- matches_clean %>%
  # Filter matches decided by penalty
  filter(PenaltyShootout == "TRUE" | PenaltyShootout == TRUE)

ggplot(penalty_matches, aes(x = as.factor(HomePenalty))) +
  geom_bar(fill = "steelblue", color = "black") +
  labs(
    title = "Distribution of Home Team Penalty Goals",
    x = "Number of Penalty Goals (Home Team)",
    y = "Frequency"
  ) +
  theme_minimal()

```



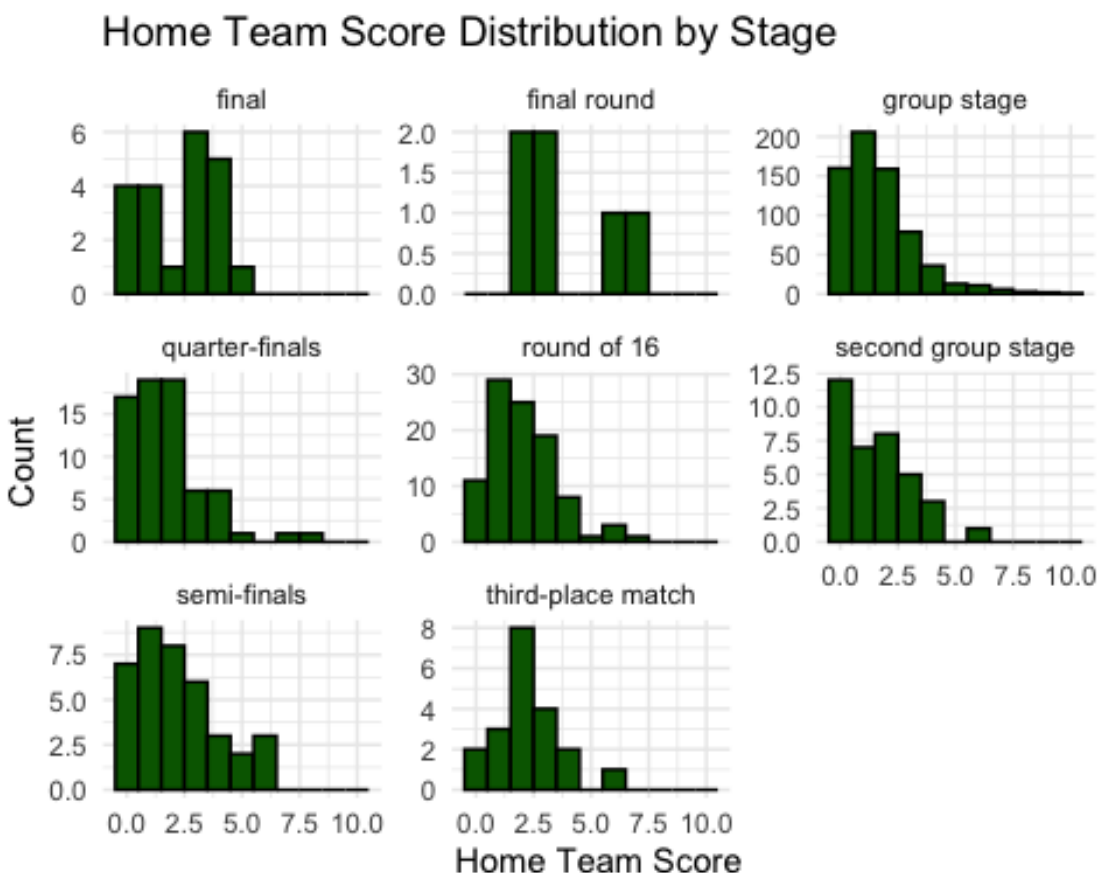
The bar chart demonstrates that the highest incidence is scoring three penalty goals, indicating a decent penalty conversion rate. The data also illustrates that the average number of successful goals is 3-4 goals per game as well. Furthermore, we see that matches in the home team scoring either just two penalty goals or even more than five penalty goals is a rare case. All of this information indicates that penalty shootouts usually run smoothly without creating any major discrepancies in the final scores or requiring a number of penalty shots.

Question 2:

The selected algorithms were conducted in accordance with the reasoning stated above, demonstrating that scores are different during the various phases of the tournaments. Histograms were plotted using the `facet_wrap()` function to ensure an optimal readability of the output. In order to aggregate scores for the individual visitors, the `case_when()` function was utilized to categorize scores in a convenient way making use of a vectorized function.

Histogram 1: Home Team Goal Distribution Chart by Stage

```
# Histogram 1: By Stage
ggplot(matches_clean, aes(x = HomeTeamScore)) +
  geom_histogram(binwidth = 1, fill = "darkgreen", color = "black") +
  facet_wrap(~Stage, scales = "free_y") +
  labs(title = "Home Team Score Distribution by Stage", x = "Home Team Score", y = "Count") +
  theme_minimal()
```



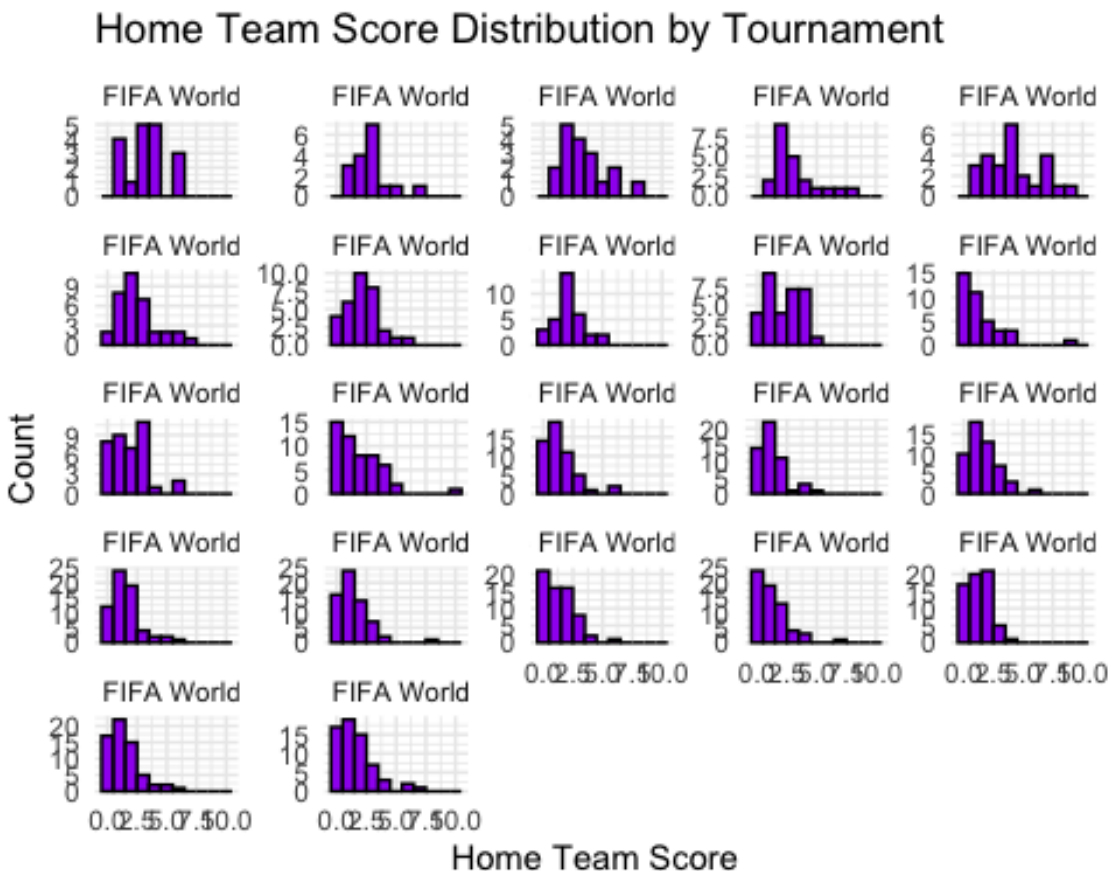
The first histogram demonstrates the variations of home team scores at different stages of a tournament. Most of the stages show a right-skewed pattern, which means that low

scores are more prevalent than high scores. The group stage has the broadest score distribution, as scoring 1 goal is the most frequent result, but there are also games where home teams score 7-10 goals. The variability in scores shows the difference in strength among teams and the offensive style of play in the early stages of the tournament. In contrast, in rounds such as quarter-finals, semi-finals, and final, the range of scores narrows considerably. Most teams score only 0-2 goals, while high-score games essentially vanish. It is likely due to the cautious and defensive playing style of teams during important matches and the fact that the remaining teams are more equally matched.

Histogram 2: Home Team Goal Distribution Chart by Tournament Name

Histogram 2: By Tournament Name

```
ggplot(matches_clean, aes(x = HomeTeamScore)) +
  geom_histogram(binwidth = 1, fill = "purple", color = "black") +
  facet_wrap(~TournamentName, scales = "free_y") +
  labs(title = "Home Team Score Distribution by Tournament", x = "Home Team
Score", y = "Count") +
  theme_minimal()
```



The second histogram shows the changes in home teams' scores during the various World Cup tournaments in history. In earlier tournaments (like 1954, 1958), the graphs have wider shapes since home teams scored more often between 3 and 6 goals indicating football was more open at that time and attacking football was preferred. However, the situation is different for modern tournaments (especially from 1990 to 2022) when home teams scored considerably less often. The highest bars are always equal to either 1 or 2 goals showing that these scores are the most common nowadays. Games with more than 3 scored goals are rather rare. The evident change in this sphere demonstrates the tendency of modern football towards defending and tactical play.

Histogram 3: Home Team Score Distribution by Away Team Score Group

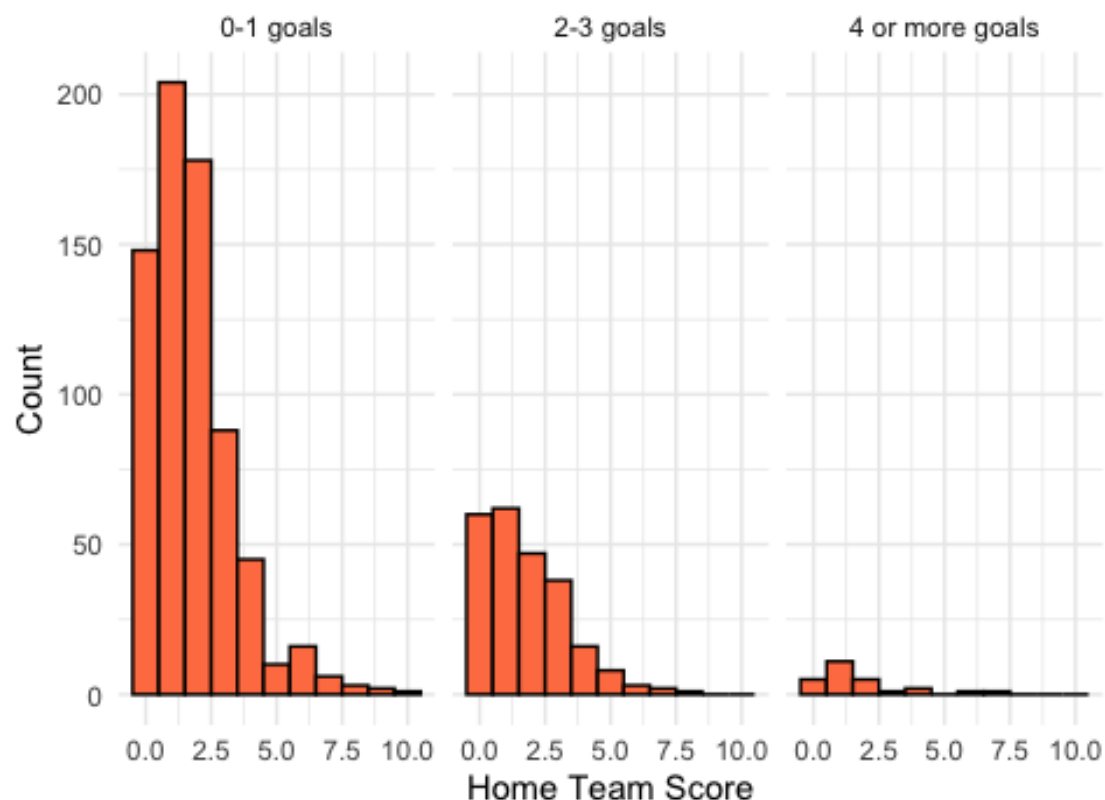
```
# Grouping AwayTeamScore
matches_grouped <- matches_clean %>%
  mutate(AwayScoreGroup = case_when(
    AwayTeamScore <= 1 ~ "0-1 goals",
    AwayTeamScore >= 2 & AwayTeamScore <= 3 ~ "2-3 goals",
    AwayTeamScore >= 4 ~ "4 or more goals",
    TRUE ~ "Unknown"
  )) %>%
  # Lock the logical order of factors for plotting
  mutate(AwayScoreGroup = factor(AwayScoreGroup, levels = c("0-1 goals", "2-3
goals", "4 or more goals")))

# Test if the grouping logic worked correctly before plotting
table(matches_grouped$AwayScoreGroup)

##
##      0-1 goals      2-3 goals 4 or more goals
##           701           237           26

# Histogram 3: By AwayTeamScore Group
ggplot(matches_grouped, aes(x = HomeTeamScore)) +
  geom_histogram(binwidth = 1, fill = "coral", color = "black") +
  facet_wrap(~AwayScoreGroup) +
  labs(title = "Home Team Score Distribution by Away Team Score Group", x =
"Home Team Score", y = "Count") +
  theme_minimal()
```

Home Team Score Distribution by Away Team Score G



The last histogram shows how the scoring of the home team varies in accordance with the number of goals scored by the away team. In games, when the away team has scored extremely low numbers of goals ('0-1 goals' group), the home team scores normally 1 or 2 goals. This is the typical scenario of the match, and the game unfolds in a simple direction when the home team gets enough chances to score. When the away team plays more proactively and scores '2-3 goals', the results with regard to the home team become more variable. The possibilities of scoring 0, 1, or 2 goals for the home team become almost equal. This means that the match is open and intense, and both teams are attacking and scoring. Finally, in the group '4 and more goals' there are very few games. Ties where the away team completely defeats the opponent and scores more than 4 goals make it impossible for the home team to resist.

Question 3:

The choice of techniques was explained with the help of the `count()` function which turned out to be useful because it allowed to perform the grouping and counting of penalty shootout appearances. The `left_join()` function was also helpful as it enabled the matching of numeric `TeamID` to their corresponding `TeamName` and `TeamCode` from the teams database.

```
# QUESTION 3
# Top 5 Home Teams
top_home_penalties <- penalty_matches %>%
  count(HomeTeamID, name = "NumberOfMatches") %>%
  arrange(desc(NumberOfMatches)) %>%
  slice_head(n = 5) %>%
  left_join(teams, by = c("HomeTeamID" = "TeamID")) %>%
  select(TeamName, TeamCode, NumberOfMatches)

# Top 5 Away Teams
top_away_penalties <- penalty_matches %>%
  count(AwayTeamID, name = "NumberOfMatches") %>%
  arrange(desc(NumberOfMatches)) %>%
  slice_head(n = 5) %>%
  left_join(teams, by = c("AwayTeamID" = "TeamID")) %>%
  select(TeamName, TeamCode, NumberOfMatches)

# Print tables
print("--- Top 5 Home Teams ---")

## [1] "--- Top 5 Home Teams ---"

print(top_home_penalties)

## # A tibble: 5 × 3
##   TeamName      TeamCode NumberOfMatches
##   <chr>         <chr>         <int>
## 1 Argentina    ARG             4
## 2 Brazil       BRA             4
## 3 Spain        ESP             4
## 4 Netherlands  NLD             3
## 5 West Germany DEU             3

print("--- Top 5 Away Teams ---")

## [1] "--- Top 5 Away Teams ---"

print(top_away_penalties)

## # A tibble: 5 × 3
##   TeamName      TeamCode NumberOfMatches
##   <chr>         <chr>         <int>
```

## 1	France	FRA	5
## 2	Argentina	ARG	3
## 3	England	ENG	3
## 4	Croatia	HRV	2
## 5	Italy	ITA	2

The statistics present the five teams that have participated the most in penalty shootouts based on their role as the home or away team. In this interesting statistic, it is important to mention that Argentina is the only team that is included in both top 5 categories according to both statistics since they were involved in 4 penalty shootouts in their home country and 3 in the opposing country when being an away team, thus demonstrating Argentina's great experience in penalty shootouts in the history of the tournament in general. Besides, France is the leader in the category of away teams according to the number of penalty shootouts since they played 5 matches as the away team. In general, the results of the top 5 list are mainly dominated by the football giants (Brazil, Spain, France, England, Italy), which is quite reasonable since the best football teams move on at the end of the tournament (in case they reach the knockout rounds) when many games can be decided by penalty shootouts.

Question 4:

The reasoning behind the choice of algorithms is that rows with NA in the Result Column have been filtered to avoid distorted graphs. In this case, both factors were shown on charts of 100% stacked position "fill". This type of chart works well for normalizing the total, giving us the possibility of comparing the proportion of wins, losses, and draws between different categories, while taking into account the number of matches in each group.

```
# Filter out matches where Result is NA for accurate modeling/visualization
```

```
analysis_data <- matches_clean %>%
```

```
  filter(!is.na(Result))
```

```
# Verify that NAs are removed and the structure is ready for plotting
```

```
str(analysis_data %>% select(ExtraTime, Stage, Result))
```

```
## tibble [964 × 3] (S3: tbl_df/tbl/data.frame)
```

```
## $ ExtraTime: Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 1 1 1 ...
```

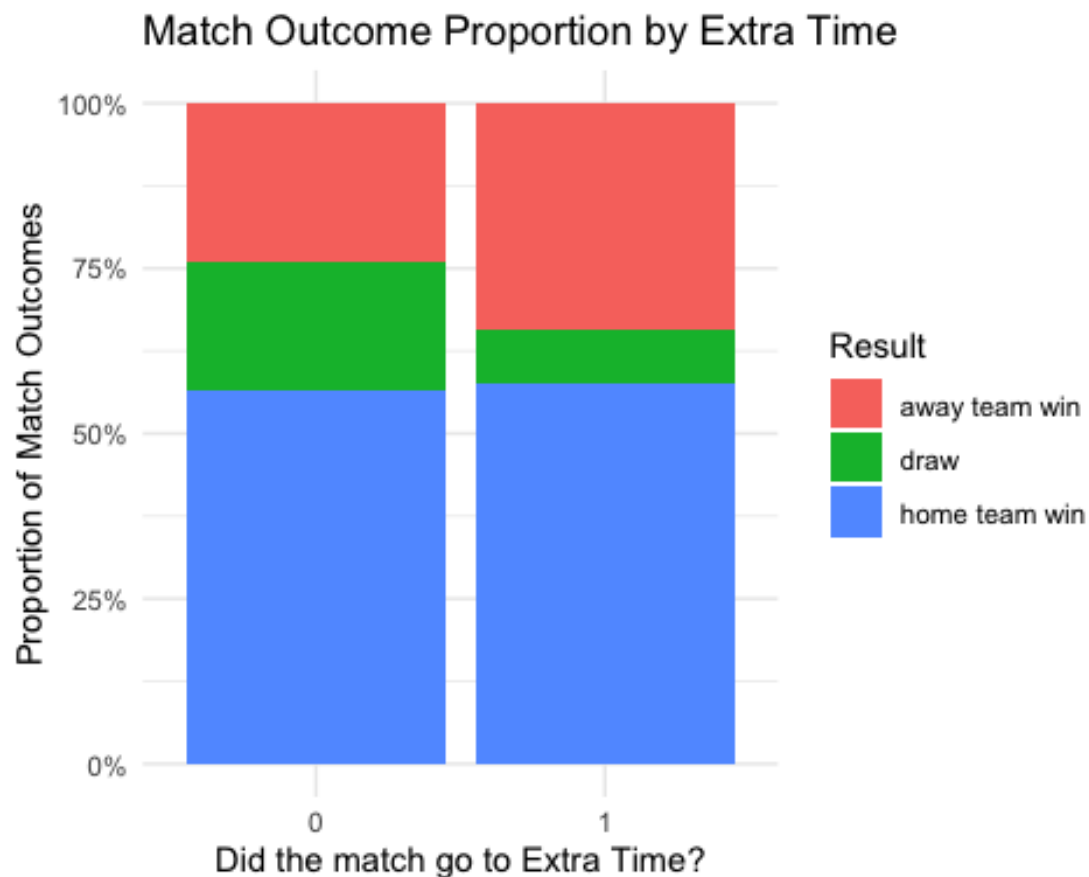
```
## $ Stage    : Factor w/ 8 levels "final","final round",...: 3 3 3 3 3 3 3 3 3 3  
3 3 ...
```

```
## $ Result   : Factor w/ 3 levels "away team win",...: 3 3 3 3 3 3 3 3 3 3
```

```
...
```

Visualizing Factor 1: Extra Time vs Result

```
# Visualizing Factor 1: Extra Time vs Result
ggplot(analysis_data, aes(x = ExtraTime, fill = Result)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = percent_format()) +
  labs(
    title = "Match Outcome Proportion by Extra Time",
    x = "Did the match go to Extra Time?",
    y = "Proportion of Match Outcomes"
  ) +
  theme_minimal()
```



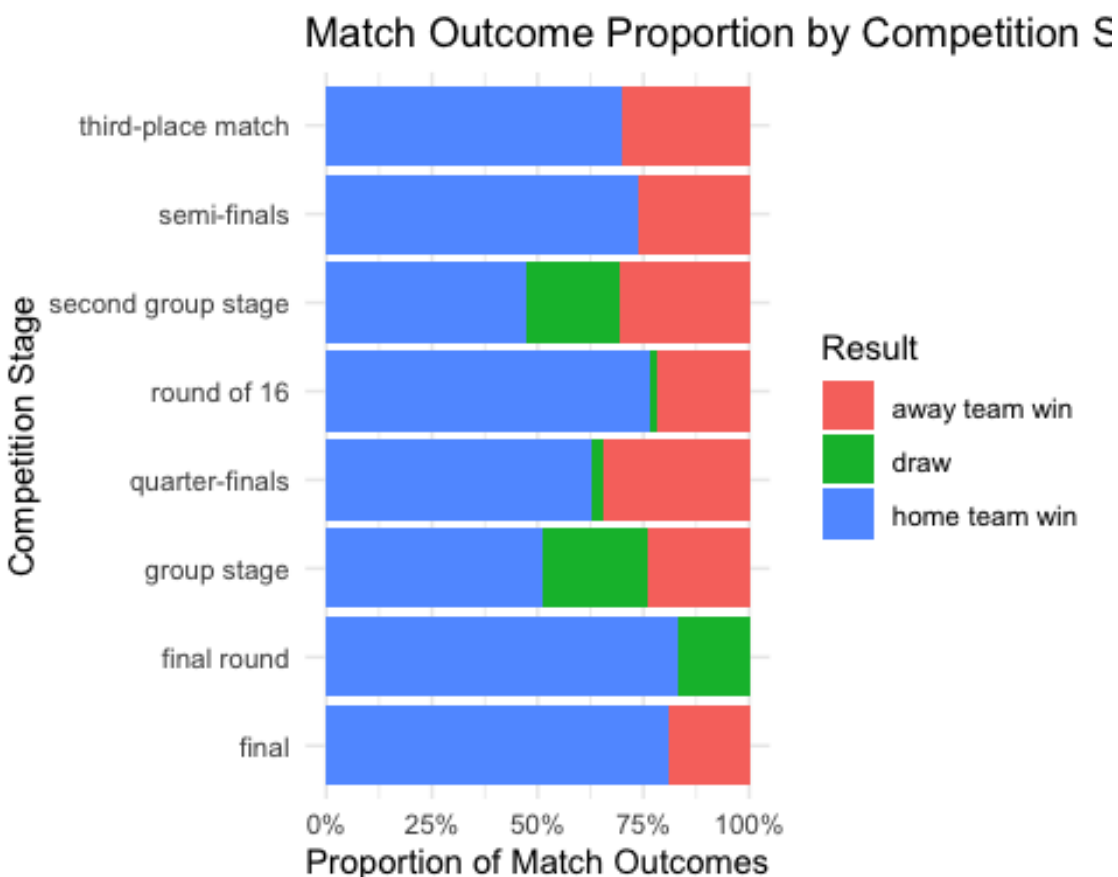
The stacked bar graph demonstrates the impact that playing Extra Time (1) instead of normal time (0) has on different match results:

- Home advantage remains unchanged: both during normal and extra times, 'home team wins' (blue bar) remains the predominant outcome, which means that the advantage of playing at home remains significant, even after playing longer than the regulation time.

- Reduction of the number of draws: One can observe a clear reduction in the 'draw' outcome (green bar) during extra time matches. This should not come as a surprise, because the aim of playing extra time in knockout rounds is unambiguously to determine the winner in the case of a tie.
- Increase of away team wins: What catches the eye is the rise in the number of matches won by the 'away team' (red bar) in matches that were played in extra time. This means that away teams must have played even better than usual during Extra Time matches.

Visualizing Factor 2: Stage vs Result

```
# Visualizing Factor 2: Stage vs Result
ggplot(analysis_data, aes(x = Stage, fill = Result)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = percent_format()) +
  coord_flip() + # Flip x and y axis for better readability of Stage names
  labs(
    title = "Match Outcome Proportion by Competition Stage",
    x = "Competition Stage",
    y = "Proportion of Match Outcomes"
  ) +
  theme_minimal()
```



This multi-layered bar graph illustrates the change in the proportion of results from matches based on the level of the tournament:

- The Presence of Draws in Group Matches: 'Group stage' and 'second group stage', the two stages with the highest proportion of draws, have the widest green bars. This is sensible, given that the teams get points for a draw in the stages and a match may finish with no clear winner.

- No Draws in Eliminating Rounds: As opposed to the knockout stages like 'final', 'semi-finals', and 'third-place match', in which a green bar does not exist at all (0%), these matches need to be won. Thus, the draws are solved at extra time or during penalty shootouts.
- Dominating Home Advantage: Throughout all stages, home team victory (represented by the blue color) is the largest part of the graph. Even under the pressure of high-pressure matches like the Round of 16 or final, the home team still gets most of the wins showing that home-field advantage is a great factor throughout the tournament.

Question 5:

My repository link:

<https://github.com/NguyenNguyet167/AP1---A2---NGUYEN-Minh-Nguyet---22218252.git>

Git commands explanation:

- **git init**: This command starts a new and blank Git project directory in the working directory. The command creates a hidden folder named `.git` that starts logging the changes made to the project.
- **git add .**: The command stages all newly modified and deleted files in the working directory. The use of the period (`.`) is indicative of the fact that all currently made changes are to be recorded for the next commit.
- **git commit -m "[message]"**: This command stores a record of the changes that have been done in the local repository. The `-m` option is used to communicate what have been changed in that particular version.
- **git remote add origin [URL]**: This command establishes a link between the local repository on the user's computer with the remote repository present on GitHub. 'Origin' is the standard name given to this connection.
- **git push -u origin main**: The command transfers the changes to the GitHub remote repository under the "main" branch. The `-u` option allows linking the current branch to the remote one for future references.
- **git checkout -b [branch_name]**: A comprehensive command used for both branch creation and checkout. It is quite useful for developing new features or testing something, e.g., a new analysis for Q4, safely in a separate branch without interfering with the stable main.

My commit history:

Commits on Jul 25, 2026

code testing for q2 & q4 NguyenNguyet167 authored now	Verified	d1d1032		<>
code testing for q2 & q4 NguyenNguyet167 committed 1 minute ago		99d57ed		<>

Commits on Jul 22, 2026

Question 4 NguyenNguyet167 authored 2 days ago	Verified	55c7423		<>
question 4 NguyenNguyet167 committed 2 days ago		a6a784c		<>
Question 3 NguyenNguyet167 authored 2 days ago	Verified	7247054		<>
question 3 NguyenNguyet167 committed 2 days ago		ec7e660		<>
Question 2 NguyenNguyet167 authored 2 days ago	Verified	20ec889		<>
question 2 NguyenNguyet167 committed 2 days ago		e671e43		<>
Question 1 NguyenNguyet167 authored 2 days ago	Verified	456df50		<>
question 1 NguyenNguyet167 committed 2 days ago		88f391e		<>

Commits on Jul 18, 2026

Delete README.md NguyenNguyet167 authored last week	Verified	4b14cb0		<>
first commit NguyenNguyet167 committed last week		50f4235		<>
Initial commit: Added datasets NguyenNguyet167 committed last week		d54217a		<>